

LABORATORY OBSERVATION

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 1

Student Name: B.Harsha Vardhan

Roll No.: A24126552075

Date: 30-08-2026

1. TITLE

Design and Implementation of College Information Portal Using HTML5

2. AIM

To develop a structured, semantic HTML5 web page for the ANITS College Information Portal that incorporates text formatting, department lists, student detail tables, interactive registration forms, and multimedia audio/video players.

3. OBJECTIVE

The objectives of this experiment are:

- To understand core HTML5 semantic tags and document layout structure.
- To implement formatted text, mathematical/chemical formulas using sup and sub tags.
- To create structured tabular data using table, tr, th, and td elements.
- To design an interactive student registration form with various HTML5 input types.
- To embed and control multimedia audio and video players.

4. THEORY

HTML5 is the standard markup language for creating modern web pages. It introduces semantic elements such as <header>, <nav>, <section>, <article>, and <footer> to structure content meaningfully.

Forms in HTML5 provide rich input controls including text, email, password, radio buttons, checkboxes, dropdown select, color pickers, and date pickers for client-side data capture.

HTML5 also provides native <audio> and <video> elements with browser controls for seamless media playback without third-party plugins.

Application Flow:

Browser -> HTML5 Document -> Semantic Elements -> Tables & Forms -> Media Players -> Rendered Webpage

5. ALGORITHM / PROCEDURE

1. Create an HTML5 document and configure the title as 'College Information Portal'.
2. Add a header section with institution title and introductory description.
3. Create a navigation bar with hyperlink anchors to sections and external portals.
4. Format about text using bold, italic, underline, superscript, and subscript tags.
5. Create ordered list for departments and unordered list for campus facilities.
6. Construct a student details table with borders, column headers, and records.
7. Build a comprehensive student registration form with text, email, password, branch select, date, color, and range inputs.

8. Embed audio and video tags with controls attribute.
9. Open in web browser and verify all elements and form controls.

6. PROGRAM

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>College Information Portal</title>
</head>
<body>

  <header>
    <h1>Welcome to Anits College</h1>
    <h2>Information Portal</h2>
    <h3>Established in 2002</h3>
    <p>Anits College is dedicated to providing world-class education in engineering and arts.</p>
  </header>

  <nav>
    <a href="#">Home</a> |
    <a href="#">About</a> |
    <a href="#">Departments</a> |
    <a href="#">Contact</a>
  </nav>
  <hr>

  <section>
    <h3>College Announcements</h3>
    <p>
      <b>Note:</b> All students must wear their <i>identity cards</i>.
      This is strictly <u>mandatory</u>.
    </p>
    <p>
      Our math formula:  $(a+b)^2 = a^2 + b^2 + 2ab$ .
      Chemical formula for Hydrogen Peroxide is  $H_2O_2$ 
    </p>

    <h3>Quick Links</h3>
    <p>
      <a href="https://www.anits.edu.in" target="_blank">This is my college website</a>
    </p>
    <p>
      <a href="https://www.google.com" target="_blank">Search in Google if you never heard my college name</a>
    </p>
    <p>
      For any queries, email us at:
      <a href="mailto:admissions@anits-college.edu">admissions@anits-college.edu</a>
    </p>

    <h3>Our Departments</h3>
    <ul>
      <li>Computer Science Engineering</li>
      <li>Electronics & Communication</li>
    </ul>
  </section>
</body>
</html>
```

```
<li>Mechanical Engineering</li>
<li>Business Administration</li>
</ul>

<h3>Admission Process (Step by Step)</h3>
<ol>
  <li>Fill the online application form</li>
  <li>Upload required documents</li>
  <li>Pay the application fee</li>
  <li>Appear for the entrance test</li>
  <li>Attend the counseling session</li>
</ol>
</section>

<section>
  <h3>Successful Students List</h3>

  <table border="1" cellpadding="10" cellspacing="0">
    <tr>
      <th>Roll No</th>
      <th>Name</th>
      <th>Branch</th>
      <th>Email</th>
    </tr>
    <tr>
      <td>101</td>
      <td>Rahul Sharma</td>
      <td>Computer Science</td>
      <td>rahul@college.edu</td>
    </tr>
    <tr>
      <td>102</td>
      <td>Priya Patel</td>
      <td>Electronics</td>
      <td>priya@college.edu</td>
    </tr>
    <tr>
      <td>103</td>
      <td>Amit Kumar</td>
      <td>Mechanical</td>
      <td>amit@college.edu</td>
    </tr>
  </table>
</section>

<article>
  <h3>Latest News</h3>
  <p>Annual college fest "TechVibes 2024" will be held on December 15th.</p>
</article>

<section>
  <h3>College Home Page Pic</h3>
  
</section>

<section>
  <h3>Student Registration Form</h3>
```

```
<form action="#" method="POST">
  <label for="name">Full Name:</label><br>
  <input type="text" id="name" name="student_name" placeholder="Enter your name" required><br>
<br>

  <label for="email">Email Address:</label><br>
  <input type="email" id="email" name="student_email" placeholder="your@email.com" required><br>
<br>

  <label for="password">Password:</label><br>
  <input type="password" id="password" name="student_password" placeholder="Create a password"
  " required><br><br>

  <label>Gender:</label><br>
  <input type="radio" id="male" name="gender" value="Male">
  <label for="male">Male</label>
  <input type="radio" id="female" name="gender" value="Female">
  <label for="female">Female</label>
  <input type="radio" id="other" name="gender" value="Other">
  <label for="other">Other</label><br><br>

  <label>Hobbies:</label><br>
  <input type="checkbox" id="reading" name="hobbies[]" value="Reading">
  <label for="reading">Reading</label>
  <input type="checkbox" id="sports" name="hobbies[]" value="Sports">
  <label for="sports">Sports</label>
  <input type="checkbox" id="music" name="hobbies[]" value="Music">
  <label for="music">Music</label>
  <input type="checkbox" id="coding" name="hobbies[]" value="Coding">
  <label for="coding">Coding</label><br><br>

  <label for="branch">Select Your Branch:</label><br>
  <select id="branch" name="branch">
    <option value="">-- Choose Branch --</option>
    <option value="CSE">Computer Science Engineering</option>
    <option value="ECE">Electronics & Communication</option>
    <option value="ME">Mechanical Engineering</option>
    <option value="CE">Civil Engineering</option>
    <option value="MBA">Business Administration</option>
  </select><br><br>

  <label for="comments">Any Additional Comments:</label><br>
  <textarea id="comments" name="comments" rows="4" cols="50" placeholder="Write your comments
  here..."></textarea><br><br>

  <label for="dob">Date of Birth:</label><br>
  <input type="date" id="dob" name="dob"><br><br>

  <label for="favcolor">Favorite Color:</label><br>
  <input type="color" id="favcolor" name="favcolor"><br><br>

  <label for="score">Satisfaction Score:</label><br>
  <input type="range" id="score" name="score" min="0" max="100" value="50"><br><br>

  <input type="submit" value="Register Now">
  <input type="reset" value="Clear All">
</form>
```

```
</section>

<section>
  <h3>College Anthem</h3>
  <audio controls>
    <source src="song.mp3" type="audio/mpeg">
    Your browser does not support audio.
  </audio>
</section>

<section>
  <h3>Campus Tour</h3>
  <video width="400" height="250" controls>
    <source src="video.mp4" type="video/mp4">
    <source src="video.webm" type="video/webm">
    Your browser does not support video.
  </video>
</section>

<footer>
  <hr>
  <p>© 2024 Anits College. All rights reserved.</p>
  <p>Contact: info@anits.edu.in | Phone: +91-12345-67890</p>
</footer>

</body>
</html>
```

7. CODE EXPLANATION

Code / Syntax	Explanation
<header>, <nav>, <footer>	Semantic HTML5 layout container elements.
<sup>, <sub>	Displays text as superscript (a²) or subscript (H₂O).
, ,	Creates ordered (numbered) and unordered (bulleted) lists.
<table>, <tr>, <th>, <td>	Structures tabular data into rows, header cells, and standard data cells.
<input type='radio' name='gender'>	Radio button group allowing only one gender selection.
<select>, <option>	Drop-down select menu for branch selection.
<audio controls>, <video controls>	Native media playback controls for audio and video files.

8. TEST CASES AND EXECUTION DATA

The experiment was executed with the following test cases to verify functionality:

Test Case	Input / Action	Expected Result	Actual Result	Status
TC-01	Page Load	Open task1.html in browser	Header, nav, and sections render cleanly	Pass
TC-02	Lists & Formulas	Verify lists and formulas	Ordered/unordered lists and formulas display properly	Pass
TC-03	Table Rendering	Check student details table	Table renders 4 columns with row borders	Pass
TC-04	Form Inputs	Fill form & click Submit	Inputs accept data, radio selection works	Pass
TC-05	Media Playback	Click play on audio/video	Media plays with default browser controls	Pass

9. OUTPUT

Output 1: Header, Quick Links & Department Lists

Welcome to Anits College

Information Portal

Established in 2002

Anits College is dedicated to providing world-class education in engineering and arts.

[Home](#) | [About](#) | [Departments](#) | [Contact](#)

College Announcements

Note: All students must wear their *identity cards*. This is strictly mandatory.

Our math formula: $(a+b)^2 = a^2 + b^2 + 2ab$. Chemical formula for Hydrogen Peroxide is H_2O_2

Quick Links

[This is my college website](#)

[Search in Google if you never heard my college name](#)

For any queries, email us at: admissions@anits-college.edu

Our Departments

- Computer Science Engineering
- Electronics & Communication
- Mechanical Engineering
- Business Administration

Admission Process (Step by Step)

1. Fill the online application form
2. Upload required documents
3. Pay the application fee
4. Appear for the entrance test
5. Attend the counseling session

Successful Students List

Roll No	Name	Branch	Email
101	Rahul Sharma	Computer Science	rahul@college.edu
102	Priya Patel	Electronics	priya@college.edu
103	Amit Kumar	Mechanical	amit@college.edu

Latest News

Annual college fest "TechVibes 2024" will be held on December 15th.

Output 2: Student Details Table & Registration Form

College Home Page Pic



Student Registration Form

Full Name:

Email Address:

Password:

Gender: ☐ Male ☐ Female ☐ Other

Hobbies: ☐ Reading ☐ Sports ☐ Music ☐ Coding

Select Your Branch:

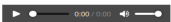
Any Additional Comments:

Date of Birth:

Favorite Color:

Satisfaction Score:

College Anthem



10. RESULT

Thus, the College Information Portal was successfully implemented using HTML5, incorporating text formatting, lists, tables, registration form inputs, and multimedia elements, and verified across all test cases.